

Figure 10.47 NOR gate equals bubbled AND gate

10.7.2 De Morgan's Second Theorem

Statement

The second theorem states that the complement of the product of two inputs is equal to the sum of its complements.

Proof

The Boolean equation for NAND gate is $Y = \overline{A.B}$.

The Boolean equation for bubbled OR gate is $Y = \overline{A} + \overline{B}$.

A and B are the inputs and Y is the output. The above two equations produces the same output for the same inputs. It can be verified by using the truth table

A	B	$A.B$	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

From the above truth table we can conclude $\overline{A.B} = \overline{A} + \overline{B}$.

Thus De Morgan's second theorem is proved. Hence, a NAND gate is equal to a bubbled OR gate.

The corresponding logic circuit diagram is shown in Figure 10.48

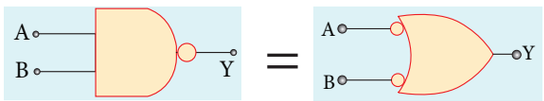


Figure 10.48 NAND gate equals bubbled OR gate

EXAMPLE 10.11

Prove the Boolean identity $AC + ABC = AC$ and give its circuit description.

Solution

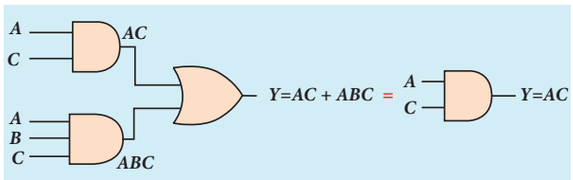
Step 1: $AC\,(1 + B) = AC.1$ [OR law-2]

Step 2: $AC\,.\,1 = AC$ [AND law – 2]

Therefore, $AC + ABC = AC$

Thus the Boolean identity is proved.

Circuit description:



10.7.3 Integrated Chips

An integrated circuit is also referred as an IC or a chip or a microchip (Figure 10.49). It consists of thousands to millions of transistors, resistors, capacitors, etc. integrated on a small flat piece of semiconductor material that is normally silicon.

Integrated circuits (ICs) are the keystone of modern electronics. With the advancement in technology and the emergence of Very Large Scale Integration (VLSI) era it is possible to fit more and more transistors on chips of same piece.

ICs have two main advantages over ordinary circuits: cost and performance. The size, speed, and capacity of chips have progressed enormously with the advancement in technology. Computers, mobile phones, and other digital home appliances are now made possible by the

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